Year 11 · AQA Level 2 Further Maths · Lesson 2

Quadratic inequalities; equations that reduce to quadratics

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Today I will learn to...

Goals

1. Solve quadratic inequalities

2. Solve equations that reduce to quadratics

Do now: retrieval practice

4 mins

1 Solve $2x + 3 > 11$

$x > 4$

2 Factorise $x^2 - x - 12$

$(x - 4)(x + 3)$

3 Solve $x^2 = 16$

$x = 4$ or $x = -4$

4 Write down the roots of $y = (x - 1)(x + 5)$

$x = 1$ and $x = -5$

Objective 1

Solve quadratic inequalities

Key idea

Learn

!

Solve the equation to find the critical values. Sketch the U-shaped curve through them. For > 0 take the parts above the x-axis (two regions). For < 0 take the part below (between the roots).

Objective 1: Solve quadratic inequalities

I do: example 1

I do

Solve $x^2 - x - 12 < 0$

1. $(x - 4)(x + 3) = 0$: critical values 4 and −3
2. Sketch a U-shape crossing the x-axis at −3 and 4
3. < 0 means below the axis: between the roots
4. $-3 < x < 4$

I do: example 2

I do

Solve $x^2 + 2x - 15 \geq 0$

1. $(x + 5)(x - 3) = 0$: $x = -5$ or $x = 3$
2. ≥ 0 means on or above the axis: outside the roots
3. $x \leq -5$ or $x \geq 3$

I do: example 3

I do

Solve $2x^2 > 5x + 3$

1. $2x^2 - 5x - 3 > 0$
2. $(2x + 1)(x - 3) = 0$: $x = -1/2$ or $x = 3$
3. > 0 means outside the roots
4. $x < -1/2$ or $x > 3$

Mini whiteboards

We do

1. Solve $x^2 - 9 \leq 0$

Answer: $-3 \leq x \leq 3$

2. Solve $x^2 - 5x + 4 > 0$

Answer: $x < 1$ or $x > 4$

3. Solve $x^2 - 2x - 8 < 0$

Answer: $-2 < x < 4$

4. Solve $x^2 \geq 25$

Answer: $x \leq -5$ or $x \geq 5$

You do: choose your challenge

You do

Bronze

Solve $x^2 - 16 > 0$

Silver

Solve $x^2 - 7x + 10 \leq 0$

Gold

Solve $x^2 + x > 6$

Platinum

Solve $3x^2 + 5x - 2 < 0$

You do: answers

Check

Bronze

$x < -4$ or $x > 4$

Silver

$2 \leq x \leq 5$

Gold

$x < -3$ or $x > 2$

Platinum

$(3x - 1)(x + 2) < 0$, so $-2 < x < 1/3$

If you finish early

Extra

Bronze

Solve $x^2 - 36 < 0$

$-6 < x < 6$

Silver

Solve $x^2 + 3x - 4 \geq 0$

$x \leq -4$ or $x \geq 1$

Gold

Solve $2x^2 - 7x + 3 < 0$

$(2x - 1)(x - 3) < 0$, so $1/2 < x < 3$

Platinum

Find the integers that satisfy $x^2 + 2x - 15 < 0$

$-5 < x < 3$: $-4, -3, -2, -1, 0, 1, 2$

Exam question 1

3 marks

Solve $x^2 + 3x - 10 < 0$

Space for working

Mark scheme: (x + 5)(x − 2) (M1). Critical values −5 and 2 (A1). −5 < x < 2 (A1)

Exam question 2

3 marks

Find the integer values of x for which $x^2 - 6x + 5 \leq 0$

Space for working

Mark scheme: (x − 1)(x − 5) ≤ 0 (M1). 1 ≤ x ≤ 5 (A1). 1, 2, 3, 4, 5 (A1)

Objective 1 achieved

Solve quadratic inequalities



Objective 2

Solve equations that reduce to quadratics

✓ Achieved: Solve quadratic inequalities

Key idea

Learn

!

Look for a hidden quadratic. In $x^4 - 5x^2 + 4 = 0$, let $y = x^2$. With fractions, multiply through by the denominator to get a quadratic. Check every answer works.

Objective 2: Solve equations that reduce to quadratics

I do: example 1

I do

Solve $x^4 - 5x^2 + 4 = 0$

1. Let $y = x^2$: $y^2 - 5y + 4 = 0$
2. $(y - 1)(y - 4) = 0$: $y = 1$ or $y = 4$
3. $x^2 = 1$ or $x^2 = 4$
4. $x = \pm 1$ or $x = \pm 2$

I do: example 2

I do

Solve $x + 6/x = 5$

1. Multiply by x: $x^2 + 6 = 5x$
2. $x^2 - 5x + 6 = 0$
3. $(x - 2)(x - 3) = 0$
4. $x = 2$ or $x = 3$

I do: example 3

I do

Solve $x - 5\sqrt{x} + 6 = 0$

1. Let $y = \sqrt{x}$: $y^2 - 5y + 6 = 0$
2. $y = 2$ or $y = 3$
3. $\sqrt{x} = 2$ or $\sqrt{x} = 3$
4. $x = 4$ or $x = 9$

Mini whiteboards

We do

1. Solve $x^4 - 10x^2 + 9 = 0$

Answer: $x = \pm 1$ or $x = \pm 3$

2. Solve $x - 4/x = 3$

Answer: $x = 4$ or $x = -1$

3. Solve $x^4 - 26x^2 + 25 = 0$

Answer: $x = \pm 1$ or $x = \pm 5$

4. Solve $x + 10/x = 7$

Answer: $x = 2$ or $x = 5$

You do: choose your challenge

You do

Bronze

Solve $x^4 - 17x^2 + 16 = 0$

Silver

Solve $x + 8/x = 6$

Gold

Solve $x - 7\sqrt{x} + 12 = 0$

Platinum

Solve $2x^4 - 3x^2 - 2 = 0$

You do: answers

Check

Bronze

$x = \pm 1$ or $x = \pm 4$

Silver

$x = 2$ or $x = 4$

Gold

$\sqrt{x} = 3$ or 4, so $x = 9$ or $x = 16$

Platinum

$(2y + 1)(y - 2) = 0$ with $y = x^2$, so $x = \pm\sqrt{2}$

If you finish early

Extra

Bronze

Solve $x^4 - 20x^2 + 64 = 0$

$x = \pm 2$ or $x = \pm 4$

Silver

Solve $x - 6/x = 1$

$x^2 - x - 6 = 0$, $x = 3$ or $x = -2$

Gold

Solve $x - 3\sqrt{x} - 10 = 0$

$\sqrt{x} = 5$ ($\sqrt{x} = -2$ is impossible), so $x = 25$

Platinum

Solve $9^x - 4 \times 3^x + 3 = 0$

Let $y = 3^x$: $(y - 1)(y - 3) = 0$, so $x = 0$ or $x = 1$

Exam question 1

3 marks

Solve $x^4 - 13x^2 + 36 = 0$

Space for working

Mark scheme: $y^2 - 13y + 36 = 0$ (M1). $y = 4$ or $y = 9$ (A1). $x = \pm 2$ or $x = \pm 3$ (A1)

Exam question 2

3 marks

Solve $2x + 3/x = 7$

Space for working

Mark scheme: $2x^2 - 7x + 3 = 0$ (M1). $(2x - 1)(x - 3) = 0$ (M1). $x = 1/2$ or $x = 3$ (A1)

Objective 2 achieved

Solve equations that reduce to quadratics



Real life

Apply

A rectangular garden has a perimeter of 30 m and its area must be more than 50 m². One side is x m. Find the possible values of x.

$x(15 - x) > 50$, so $x^2 - 15x + 50 < 0$, $(x - 5)(x - 10) < 0$ and $5 < x < 10$

Extension: stretch yourself

Stretch

1. Solve $x^2 + 4 > 4x$

$(x - 2)^2 > 0$: every x except $x = 2$

2. Solve $(x^2 - 3)^2 - 4(x^2 - 3) + 3 = 0$

Let $y = x^2 - 3$: $(y - 1)(y - 3) = 0$, so $x^2 = 4$ or 6 and $x = \pm 2$ or $\pm\sqrt{6}$

Exit ticket: exam questions

Exit

1. Solve $x^2 - 3x - 10 < 0$ **(2 marks)**

2. Solve $x^2 - 4x \geq 0$ **(2 marks)**

3. Solve $x^4 - 5x^2 - 36 = 0$ **(2 marks)**

Exit ticket: mark scheme

Mark it

Q1 $(x - 5)(x + 2)$ (M1). $-2 < x < 5$ (A1) [2]

Q2 $x(x - 4)$ (M1). $x \leq 0$ or $x \geq 4$ (A1) [2]

Q3 $(x^2 - 9)(x^2 + 4) = 0$ (M1). $x = 3$ or $x = -3$ (A1) [2]

Total: ____ / 6

Today I achieved

- ✔ **Solve quadratic inequalities**
- ✔ **Solve equations that reduce to quadratics**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Solve $x^2 - 25 < 0$ [1]

2. Solve $x^2 + 4x - 12 \geq 0$ [2]

3. Solve $2x^2 + x - 6 > 0$ [3]

4. Solve $x^4 - 29x^2 + 100 = 0$ [3]

5. Solve $x + 12/x = 7$ [2]

Extra practice: Mr Barton Maths: AQA Further Maths → search "Quadratic inequalities"